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Volume 2....No. 1, part 2, War, parts 1 and 2.
Volume 3....No. 1, part 5, Interior, parts 1 and 2.
Volume 4....No. 1, parts 3, 4, 6, and 7, and No. 2.
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Detroit and Duluth—difference of longitude.

Date.	From Duluth record.	From Detroit record.	Mean.	Wave and armature time.
	m. *s.*	*m.* *s.*	*m.* *s.*	*s.*
June 15, 1871	36 05.98	36 05.70	36 05.84	0.140
June 26, 1871	36 05.96	36 05.70	36 05.83	0.130
July 11, 1871	36 05.92	36 05.66	36 05.79	0.130

Duluth and Saint Paul—direct comparison of clock and mean-time chronometer, being at Duluth, for each comparison, the mean of 79 consecutive second-breaks of the mean-time chronometer as recorded on the chronograph, and at Saint Paul coincidences of clock and chronometer.

Date.	Duluth chronograph.		Coincidences at Saint Paul.	
	Clock No. 256.	Chron. No. 4212.	Clock No. 256.	Chron. No. 4212.
	h. *m.* *s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*h.* *m.* *s.*
June 14, 1871	18 56 49.82	= 13 22 40.0	18 36 13.0	= 13 02 06.5
	19 03 00.78	= 13 28 50.0	18 42 16.0	= 13 08 08.5
June 15, 1871	18 46 49.15	= 13 08 50.0	18 30 12.0	= 12 52 15.5
			18 33 02.0	= 12 55 05.0
			18 36 08.0	= 12 58 10.5
			18 39 06.0	= 13 01 08.0

The same as above, with errors of clock applied.

Date.	Duluth chronograph.		Coincidences at Saint Paul.	
	Clock No. 256.	Chron. No. 4212.	Clock No. 256.	Chron. No. 4212.
	h. *m.* *s.*	*h.* *m.* *s.*	*h.* *m.* *s.*	*h.* *m.* *s.*
June 14, 1871	18 56 11.23	= 18 52 07.35	18 35 34.41	= 18 31 30.58
	19 02 22.19	= 18 58 19.32	18 41 37.41	= 18 37 33.53
June 15, 1871	18 46 10.81	= 18 42 07.17	18 29 33.66	= 18 25 30.04
			18 32 23.66	= 18 28 19.99
			18 35 29.66	= 18 31 25.98
			18 38 27.66	= 18 34 23.95

Duluth and Saint Paul—difference of longitude.

Date.	From Duluth record.	From Saint Paul coincidences.	Mean.
	m. *s.*	*m.* *s.*	*m.* *s.*
June 14, 1871	4 03.88	4 03.83	
	4 03.87	4 03.88	4 03.86
June 15, 1871	4 03.64	4 03.62	
		4 03.67	4 03.66
		4 03.68	
		4 03.71	

Observations for relative personal equation were made by Assistants O. B. Wheeler and G. Y. Wisner on April 22 and 24 at Detroit. These observations were finally discarded, owing to the long interval between them and the observations for longitude, and also on account of later observations when the observers had had more constant practice. Observations were also made at Detroit by General C. B. Comstock and Assistant O. B. Wheeler on May 6, 9, and 11, July 7 and 9, and August 3 and 4, with the following results:

Personal equation, General C. B. Comstock and Assistant O. B. Wheeler.

May 6, 1871, C. B. C. — O. B. W ..... = + 0s.020 ± 0s.033
May 9, 1871, C. B. C. — O. B. W ..... = + 0s.018 ± 0s.029
May 11, 1871, C. B. C.— O. B. W ..... = + 0s.016 ± 0s.018

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Messrs. Eisenmann and Metcalf. The reduction of personal equation has been by Messrs. Russell and Eisenmann. I have given close attention to the computations, and have compared and combined the results.

Very respectfully, your obedient servant,

O. B. WHEELER,
Assistant Lake Survey.

Major C. B. COMSTOCK,
United States Engineers, Brevet Brigadier-General, U. S. A.,
In charge United States Lake Survey.

Time-determination tables.

In the following are contained the abbreviations—
C Δi = Reduction to the middle wire;
Aberr. = Diurnal aberration;
C c = Collimation-correction;
B b = Level-correction;
t' = Observed time of transit of a star, commonly the mean of nine wires;
$t = t'$, the above correction being made;
α = Right ascension of a star;
$(\alpha - t) - (\text{an assumed } \Delta t)$ = The absolute term of the equations of condition;
a = Deviation from the meridian, + when the instrument points east of south;
c = Distance of middle wire from the line of collimation;
ρ = Hourly rate of clock;
Δt = Clock-correction — when fast; + when slow;
Where C Δi is omitted, t' = the mean of the several wires reduced to the middle wire.

DETROIT, *June* 15, 1871.
Latitude, 42° 20′ +. Observer, Assistant O. B. Wheeler.

Ill.	Star.	CΔi	Aberr.	Rate.	Aa	Bb	Cc	t'	α	Δt	v
		s.	s.	s.	s.	s.	s.	h. m. s.	h. m. s.	m. s.	s.
E.	α Bootis	− .06	− .01	+ .32	− .04	+ .10	− .16	14 11 12.54	14 09 47.81	−1 24.88	+ .01
	β Libræ	− .06	− .01	+ .23	− .07	+ .10	− .16	15 11 30.15	15 10 05 49	24.69	− .18
	μ^1 Bootis	− .08	− .02	+ .21	− .01	+ .21	− .20	21 03.70	19 38.71	25.10	+ .33
	α Coronæ Borealis	− .07	− .02	+ .20	− .02	+ .19	− .18	30 40.03	29 15.14	24.99	+ .12
	ϵ Serpentis	− .06	− .01	+ .18	− .05	+ .16	− .16	45 49.44	44 24.75	24.75	− .12
	δ Ophiuchi	− .06	− .01	+ .14	− .06	+ .15	− .16	16 09 01.47	16 07 36.79	24.68	− .19
	Ophiuchi	− .06	− .01	+ .11	− .07	+ .15	− .16	31 29.72	30 05.11	24.57	− .30
	κ Ophiuchi	− .06	− .01	+ .08	− .05	+ .22	− .16	53 00.25	51 35.38	24.89	+ .01
E.	ϵ Ursæ Minoris	.....	− .10	+ .07	+ .40	+1.49	− .30	17 00 48.86	59 25.56	.....	.....
W.	α' Herculis	+ .06	− .01	+ .05	− .09	+ .27	+ .17	10 12.28	17 08 47.63	25.10	+ .23
	α Ophiuchi	+ .06	− .01	+ .02	− .09	+ .27	+ .16	30 23.13	28 58.48	25.06	+ .19
	γ Draconis	+ .10	− .02	− .01	+ .02	+ .47	+ .26	55 03.31	53 39.12	25.01	+ .14
	d Sagittarii	+ .06	− .01	− .12	− .16	+ .10	+ .17	19 11 31.64	19 10 06.86	24.82	− .05
	ω Aquilæ	+ .06	− .01	− .13	− .09	+ .17	+ .16	13 12.06	11 47.22	25.00	+ .13
	δ Aquilæ	+ .06	− .01	− .14	− .11	+ .15	+ .16	20 25.94	19 01.15	24.90	+ .03
	γ Aquilæ	+ .06	− .01	− .17	− .10	+ .16	+ .16	41 34.02	40 09.10	25.02	+ .15
	α Aquilæ	+ .06	− .01	− .17	− .10	+ .13	+ .16	45 55.54	44 30.83	24.78	− .11
	β Aquilæ	+ .06	− .01	− .18	− .10	+ .15	+ .16	50 24.92	49 00.04	24.96	+ .09
W.	λ Ursæ Minoris	.....	− .74	− .19	+6.76	+6.58	+8.31	55 17.45	54 13.19	.....	.....
E.	λ Ursæ Minoris	.....	− .74	.....	.....	+6.88	−8.31	33.75	.....	.....	.....
	α^2 Capricorni	− .06	− .01	− .21	− .15	.....	− .17	20 12 20.21	20 10 55.17	− 24.45	− .42

Normal equations.

$$+ 24.11a \quad + 1.60\rho - 2.07\,\delta\theta + 0.00 = 0$$
$$+ 1482.91a' + 73.87\rho - 32.93\,\delta\theta + 222.45 = 0$$
$$+ 1.60a + 73.87a' + 61.27\rho \quad + 7.75 = 0$$
$$- 2.07a - 32.93a' \quad + 18.00\,\delta\theta + 10.50 = 0$$

$a = -0^s.084 \qquad a' = -0^s.176 \qquad c = -0^s.16 \qquad \rho = +0^s.088$

$\Delta t = -1^m.24^s.87 \pm 0^s.030$ at $17^h.47^m$.

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DETROIT, *June* 26, 1871.

Latitude, 42° 20′ +. Observer, Assistant O. B. Wheeler.

Ill.	Star.	CΔi	Aberr.	Rate.	Aa	Bb	Cc	t′	a	Δt
		s.	s.	s.	s.	s.	s.	h. m. s.	h. m. s.	m. s.
E.	β Bootis	− .08	− .02	− .16	+ .01	+ .15	− .10	14 58 35.55	14 57 06.66	−1 28.89
	48 Cephei, L. C	+ .14	+ .06	− .16	+ .39	− .27	+ .35	15 05 28.01	3 03 59.65	
	μ^1 Bootis	− .08	− .02	− .14	+ .01	+ .18	− .10	21 07.49	19 38.63	
	α Coronæ Borealis	− .07	− .02	− .14	+ .03	+ .17	− .09	30 43.75	29 15.09	
	α Serpentis	− .06	− .01	− .13	+ .06	+ .13	− .08	39 24.93	37 56.37	
	ε Serpentis	− .06	− .01	− .13	+ .06	+ .14	− .08	45 53.16	44 24.74	
	ε Coronæ Borealis	− .07	− .02	− .11	+ .03	+ .20	− .09	53 45.14	52 16.50	
	δ Ophiuchi	− .06	− .01	− .10	+ .07	+ .14	− .08	16 09 05.23	16 07 36.81	
	ξ Ophiuchi	− .06	− .01	− .08	+ .08	+ .13	− .08	31 33.49	30 05.14	
E.	κ Ophiuchi	− .06	− .01	− .06	+ .06	+ .18	− .08	53 03.93	51 35.42	
W.	α Ophiuchi	+ .06	− .01	− .03	+ .05	+ .15	− .08	17 30 26.85	17 28 58.56	
W.	ξ Aquilæ	+ .06	− .01	+ .05	+ .05	+ .20	− .08	19 00 58.79	18 59 30.57	
	ω Aquilæ	+ .06	− .01	+ .06	+ .05	+ .16	− .08	13 15.69	19 11 47.40	
	δ Aquilæ	+ .06	− .01	+ .07	+ .06	+ .13	− .08	21 29.48	19 01.35	
	κ Aquilæ	+ .06	− .01	+ .08	+ .08	+ .10	− .08	31 26.91	29 58.78	
W.	λ Ursæ Minoris	.....	− .74	+ .10	−3.85	+4.78	−4.12	55 40.46	54 16.37	
E.	λ Ursæ Minoris	.....	− .74	.....	.....	+4.78	−4.12	48.70	.....	
	ε Delphini	− .06	− .01	+ .13	+ .05	+ .11	− .08	20 28 32.83	20 27 04.55	
	α Cygni	− .08	− .02	+ .14	− .01	+ .18	− .11	38 32.63	37 03.96	
	μ Aquarii	− .06	− .01	+ .14	+ .08	+ .08	− .08	47 11.52	45 43.19	
	32 Vulpeculæ	− .07	− .02	+ .15	+ .03	+ .12	− .09	50 33.70	49 05.35	
	61[1] Cygni	− .08	− .02	+ .16	+ .01	+ .16	− .10	21 02 37.45	21 01 08.83	
E.	ζ Cygni	− .07	− .02	+ .16	+ .03	+ .13	− .09	08 56.69	07 28.32	−1 28.69

Normal equations.

$$+ 1502.25a + 83.30\rho - 25.67\,\delta\theta - 157.33 = 0$$
$$+ 83.30a + 111.16\rho - 2.30 = 0$$
$$- 25.67a + 22.00\,\delta\theta + 14.86 = 0$$
$$a = + 0s.10 \qquad \rho = - 0s.053 \qquad c = - 0s.08$$
$$\Delta t = - 1m.\,28s.55 \pm 0s.020 \text{ at } 18h.\,04m.$$

DETROIT, *July* 11, 1871.

Latitude, 40° 20′ +. Observer, Assistant O. B. Wheeler.

Ill.	Star.	CΔi	Aberr.	Rate.	Aa	Bb	Cc	t′	a	Δt
		s.	s.	s.	s.	s.	s.	h. m. s.	h. m. s.	m. s.
E.	ε Coronæ Borealis	−.07	−.02	−.21	− .02	+ .29	+ .08	15 53 20.56	15 52 16.39	−1 04.28
	β′ Scorpii	−.06	−.01	−.20	− .06	+ .14	+ .07	59 02.10	57 57.96	
	δ Ophiuchi	−.06	−.01	−.19	− .04	+ .19	+ .07	16 08 40.84	16 07 36.77	
	ξ Ophiuchi	−.06	−.01	−.16	− .05	+ .16	+ .07	31 09.15	30 05.13	
	κ Ophiuchi	−.06	−.01	−.13	− .03	+ .23	+ .07	52 39.50	51 35.40	
E.	ε Ursæ Minoris	.....	−.10	−.12	+ .29	+ 1.54	+ .53	17 00 25.84	59 23.51	
W.	ε Ursæ Minoris	.....	−.10	.....	+ .29	+ 1.87	− .53	26.56	.....	
	α′ Herculis	+.06	−.01	−.10	− .03	+ .32	− .07	17 09 51.59	17 08 47.68	
	α Ophiuchi	+.06	−.01	−.08	− .03	+ .33	− .07	30 02.50	28 58.39	
	μ Herculis	+.07	−.02	−.06	− .02	+ .43	− .08	42 31.33	41 26.49	
	γ Draconis	+.10	−.02	−.04	+ .01	+ .67	− .11	54 42.51	53 39.11	
	δ Ursæ Minoris	.....	−.24	−.01	+ .71	+ 4.56	−1.18	18 15 15.15	14 15.31	
W.	α Lyræ	+.08	−.02	+ .01	− .01	+ .49	− .09	33 39.92	32 36.56	
W.	κ Aquilæ	+.06	−.01	+.09	− .04	+ .25	− .07	19 31 02.69	19 29 59.60	
	γ Aquilæ	+.06	−.01	+.11	− .03	+ .33	− .07	41 13.13	40 09.51	
	α Aquilæ	+.06	−.01	+.11	− .03	+ .32	− .07	45 34.85	44 31.56	
W.	λ Ursæ Minoris	.....	−.74	+.13	+2.31	+13.16	−3.72	5 10.38	54 17.36	
E.	λ Ursæ Minoris	.....	−.74	.....	+2.31	+ 9.45	+3.72	06.65	.....	
	ε Delphini	−.06	−.01	+.17	− .03	+ .25	+ .07	20 28 08.42	27 04.83	
	μ Aquarii	−.06	−.01	+.20	− .05	+ .20	+ .07	46 47.11	45 43.51	
	61[1] Cygni	−.08	−.02	+.22	− .01	+ .41	+ .09	21 02 12 96	21 01 09.17	
E.	ζ Cygni	−.07	−.02	+.23	− .02	+ .33	+ .08	08 32.38	07 28.65	

Normal equations.

$$+ 1652.45a + 52.04\rho - 46.98\,\delta\theta + 50.36 = 0$$
$$+ 52.04a + 59.58\rho + 8.07 = 0$$
$$- 46.98a + 20.00\,\delta\theta + 18.99 = 0$$
$$a = - 0s.06 \qquad c = + 0s.07 \qquad \rho = - 0s.084$$
$$\Delta t = - 1m.\,04s.08 \pm 0s.021 \text{ at } 18h.\,24m.$$

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DULUTH, *July* 11, 1871.

Latitude, 46° 45′ +. Observer, G. Y. Wisner.

Ill.	Star.	CΔi	Aberr.	Aa	Bb	Cc	t'	α	Δt	r
		s.	s.	s.	s.	s.	h. m. s.	h. m. s.	s.	s.
W.	ϵ Bootis	+ .01	− .02	+ .08	− .05	− .01	14 39 50.69	14 39 22.30	−28.40	−.01
	β Bootis	+ .01	− .02	+ .03	− .07	− .01	57 34.90	57 06.43	28.40	−.01
	β Libræ	+ .01	− .01	+ .18	− .03	− .01	15 10 33.63	15 10 05.37	28.40	−.03
	μ^1 Bootis	+ .01	− .02	+ .04	− .08	− .01	20 06.89	19 38.43	28.40	−.01
	α Coronæ Borealis	+ .01	− .02	+ .08	− .06	− .01	29 43.25	29 14.96	28.29	−.13
	α Serpentis	+ .01	− .01	+ .14	− .04	− .01	38 24.60	37 56.28	28.41	−.07
	ϵ Serpentis	+ .01	− .01	+ .14	− .04	− .01	44 52.94	44 24.66	28.37	−.04
	ε Coronæ Borealis	+ .01	− .02	+ .08	− .05	− .01	52 44.65	52 16.39	28.37	−.15
	δ Ophiuchi	+ .01	− .01	+ .16	− .03	− .01	16 08 05.05	16 07 36.77	28.40	−.01
	τ Herculis	+ .01	− .02	.00	− .06	− .01	16 22.11	15 53.49	28.54	−.13
	ζ Ophiuchi	+ .01	− .01	+ .18	− .03	− .01	30 33.52	30 05.13	28.53	−.12
	ζ Herculis	+ .01	− .02	+ .07	− .07	− .01	36 55.54	36 27.12	28.40	−.01
	η Herculis	+ .01	− .02	+ .04	− .10	− .01	38 58.69	38 30.21	28.40	−.01
W.	ϵ Ursæ Minoris	.....	− .10	− .92	− .53	− .10	59 53.55	16 59 23.51	.....	.....
E.	ϵ Ursæ Minoris	.....	− .10	.....	−1.10	+ .10	53.91	.....	.....	.....
	α^1 Herculis	− .01	− .01	+ .12	− .16	+ .01	17 09 16.27	17 08 47.68	28.54	−.13
	α Ophiuchi	− .01	− .01	+ .12	− .16	+ .01	29 27.00	28 58.59	28.36	−.05
E.	μ Herculis	.00	− .02	+ .08	− .19	+ .01	41 54.79	41 26.49	28.18	−.34
E.	ζ Aquilæ	− .01	− .01	+ .04	+ .17	+ .01	18 59 58.95	18 59 30.72	28.43	+.02
	ω Aquilæ	− .01	− .01	+ .04	+ .16	+ .01	19 12 15.85	19 11 47.58	28.46	−.05
	δ Aquilæ	− .01	− .01	+ .04	+ .14	+ .01	19 29.81	19 01.54	28.44	+.03
	κ Aquilæ	− .01	− .01	+ .05	+ .12	+ .01	30 27.33	29 59.00	28.49	+.01
	γ Aquilæ	− .01	− .01	+ .04	+ .18	+ .01	40 37.77	40 09.51	28.47	−.01
	α Aquilæ	− .01	− .01	+ .04	+ .17	+ .01	44 43.39	44 31.26	28.33	−.15
E.	λ Ursæ Minoris	.....	− .74	−2.28	+7.67	.....	54 41.13	54 17.36	.....	.....
W.	λ Ursæ Minoris	.....	− .74	.....	+7.78	.....	41.05	.....	.....	.....
	α^2 Capricorni	+ .01	− .01	+ .06	+ .10	− .01	20 11 24.05	20 10 55.70	28.50	−.09
	ϵ Delphini	+ .01	− .01	+ .04	+ .07	− .01	27 33.25	27 04.83	28.52	−.11
	α Cygni	+ .01	− .02	.00	+ .07	− .01	37 32.63	37 04.25	28.43	−.02
	μ Aquarii	+ .01	− .01	+ .05	+ .01	− .01	46 11.95	45 43.52	28.48	−.07
	32 Vulpeculæ	+ .01	− .02	+ .02	+ .01	− .01	49 34.12	49 05.65	28.48	−.07
	ν Cygni	+ .01	− .02	+ .01	+ .02	− .01	52 52.33	52 23.91	28.43	−.02
	61[1] Cygni	+ .01	− .02	+ .01	+ .02	− .01	21 01 37.50	21 01 09.17	28.34	−.07
	ξ Cygni	+ .01	− .02	+ .02	+ .01	− .01	07 57.11	07 28.65	28.47	−.06
	1 Pegasi	+ .01	− .02	+ .03	+ .01	− .01	16 37.39	16 09.16	28.31	−.10
	β Aquarii	+ .01	− .01	+ .05	+ .01	− .01	25 16.03	24 47.76	28.32	−.09
W.	ϵ Pegasi	+ .01	− .01	+ .04	+ .01	− .01	38 21.04	37 52.71	28.37	−.04

Normal equations.

$$+22.86a \qquad + 2.92\,\delta\theta - 3.70 = 0$$
$$+1275.52a' - 26.41\,\delta\theta - 92.59 = 0$$
$$+2.92a - 26.41a' + 35.00\,\delta\theta + 15.46 = 0$$

$$= +0s.01 \qquad a = +0s.21 \qquad a' = +0s.06$$

$$\Delta t = -28s.41 \pm 0s.010 \text{ at } 18h.\ 30m.$$

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